6TH GRADE MATH · UNIT 6: GEOMETRY

Dream Bedroom Makeover

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students redesign a bedroom. They decompose floor plans and rugs into rectangles and triangles, use nets to find wall and box surface areas, and compute volumes of storage, then price the whole makeover.

The budget gives every measurement a purpose and lets students compare design choices.

Standards Addressed

- 6.G.A.1 Area of triangles, quadrilaterals, and polygons by composing and decomposing
- 6.G.A.2 Volume of right rectangular prisms with fractional edges
- 6.G.A.3 Polygons in the coordinate plane
- 6.G.A.4 Nets and surface area

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: show a messy bedroom photo. Part 1 floor area and rugs.
Day 2	Part 2 wall area and nets; build a box net from paper.
Day 3	Part 3 storage volume and budget; create the design board.

Materials

- Student handout and Price Cards
- Grid paper and rulers
- Scissors and cardstock for box nets
- Calculators

How the Three Versions Differ

Level	What changes
CP2	Rectangles, simple triangles, four-wall area, one box volume, and a short budget.
CP1	Composite floors, trapezoid and triangle rugs, subtracting doors and windows, fitting boxes on a shelf.
Honors	Mixed flooring, coordinate polygons, comparing box designs by surface area, and a budget pitch with percent used.

Watch For (Common Misconceptions)

- Forgetting the $\frac{1}{2}$ in the triangle area formula.
- Counting only three faces of a prism.
- Assuming boxes fit just because the volume is large enough.

Extension Ideas

- Build a scale model of the room with a shoebox.
- Price real flooring from a store website.
- Design a loft bed and find the volume under it.

Answer Key – CP2

Part 1: Floor Plan

#	Task	Answer
1	Find the area of the floor.	12 × 10 = 120 sq ft
2	Find the cost to cover the floor with carpet.	120 × $2.25 = $270
3	A triangle rug has a base of 6 ft and a height of 4 ft. Find its area.	$\frac{1}{2}$ × 6 × 4 = 12 sq ft
4	An L-shaped desk top is made of a 5 ft by 2 ft rectangle and a 3 ft by 2 ft rectangle. Find its area.	10 + 6 = 16 sq ft

Part 2: Paint the Walls

#	Task	Answer
5	Find the area of one long wall (12 ft by 8 ft) and one short wall (10 ft by 8 ft).	96 sq ft and 80 sq ft
6	Find the total area of all four walls.	2(96) + 2(80) = 352 sq ft
7	One gallon of paint covers 350 sq ft. How many gallons do you need? What does it cost?	352 > 350, so 2 gallons; 2 × $32 = $64
8	Draw the net of a storage box (rectangular prism) 2 ft by 1.5 ft by 1 ft.	Net of six rectangles: two 2 × 1.5, two 2 × 1, two 1.5 × 1.

Part 3: Storage & Budget

#	Task	Answer
9	Find the volume of one storage box.	2 × 1.5 × 1 = 3 cubic ft
10	You buy 3 boxes. What is their total volume and total cost?	9 cubic ft; 3 × $18 = $54
11	Add up your carpet, paint, rug ($4 per sq ft), and boxes. Is it under the $900 budget?	270 + 64 + 48 + 54 = $436; yes, under $900.

Reflection

#	Prompt	Look for
R1	Why is area measured in square feet but volume in cubic feet?	Area counts squares that cover a flat surface; volume counts cubes that fill space.
R2	How did the net help you find surface area?	It shows every face flat, so you can add the area of each rectangle.

Answer Key – CP1

Part 1: Flooring Choices

#	Task	Answer
1	Find the total floor area.	120 + 12 = 132 sq ft
2	Compare the cost of wood flooring and carpet for the whole floor. How much more is wood?	Wood 132 × 3.50 = $462; carpet 132 × 2.25 = $297; wood costs $165 more.
3	A rug is a trapezoid with bases 8 ft and 5 ft and height 6 ft. Find its area and cost.	$\frac{1}{2}(8 + 5)(6) = 39$ sq ft; 39 × $4 = $156
4	A triangular wall decal has base 18 in and height 12 in. Find its area and cost.	$\frac{1}{2}(18)(12) = 108$ sq in; 108 × 0.75 = $81

Part 2: Paint & Surfaces

#	Task	Answer
5	Find the area of the four walls of the 12 ft by 10 ft room.	2(12 × 8) + 2(10 × 8) = 352 sq ft
6	Subtract the door and windows. How much wall area will be painted?	352 − 21 − 2(14) = 303 sq ft
7	Two coats are needed. How many gallons should you buy, and what is the cost?	606 ÷ 350 ≈ 1.73 → 2 gallons = $64
8	Find the surface area of one storage box (2 ft × 1.5 ft × 1 ft) using a net. How much fabric covers 4 boxes?	2(3) + 2(2) + 2(1.5) = 13 sq ft; 52 sq ft for 4

Part 3: Storage & Final Budget

#	Task	Answer
9	A closet shelf is 4 ft by $1\frac{1}{2}$ ft by $2\frac{1}{2}$ ft of space. How many storage boxes (2 × 1.5 × 1 ft) fit by volume? Can they actually fit?	Shelf volume 4 × 1.5 × 2.5 = 15 cubic ft; 15 ÷ 3 = 5 boxes by volume. Arranged: 2 boxes across (4 ft) × 1 deep × 2 high = 4 boxes fit.
10	Find the cost of the boxes that fit.	4 × $18 = $72
11	Choose flooring, a rug, paint, decals, and boxes. Find the total and show it is under $900.	Answers vary; e.g., carpet 297 + rug 156 + paint 64 + decal 81 + boxes 72 = $670.

Reflection

#	Prompt	Look for
R1	Why can the number of boxes that fit by volume be more than the number that actually fit?	Volume ignores shape; boxes cannot be cut, so empty gaps remain.

#	Prompt	Look for
R2	How did decomposing shapes help you find areas?	Breaking a shape into rectangles and triangles lets you use formulas you know and add the parts.

Answer Key – Honors

Part 1: Floor & Rug Design

#	Task	Answer
1	The floor will be wood except a square carpeted reading nook 5 ft on a side. Find the cost of the floor.	Total 132 sq ft. Carpet 25 × 2.25 = $56.25. Wood 107 × 3.50 = $374.50. Total $430.75.
2	A hexagon rug is made of 6 congruent triangles, each with base 3 ft and height 2.6 ft. Find its area and cost.	$6 \times \frac{1}{2}(3)(2.6) = 23.4$ sq ft; 23.4 × 4 = $93.60
3	Plot the rug corners on a grid: (2, 1), (8, 1), (10, 5), (2, 5) (1 unit = 1 ft). Find its area by decomposing.	Rectangle (2,1)-(8,5): 6 × 4 = 24; triangle (8,1),(10,5),(8,5): $\frac{1}{2}$(2)(4) = 4; total 28 sq ft
4	Which rug gives more area per dollar at $4 per sq ft: hexagon or the trapezoid-shaped rug from the grid? Explain.	Both cost $4 per sq ft, so area per dollar is the same (0.25 sq ft per dollar); the grid rug is larger.

Part 2: Paint & Box Design

#	Task	Answer
5	Find the gallons and cost of paint for the walls AND the ceiling (12 × 10 ft) with two coats.	Walls 303 + ceiling 120 = 423 sq ft × 2 = 846 → 846 ÷ 350 ≈ 2.42 → 3 gallons = $96
6	Two box designs hold 3 cubic ft: A is 2 × 1.5 × 1 ft, B is 1.5 × 1.5 × $1\frac{1}{3}$ ft. Which uses less cardboard?	A: 13 sq ft. B: 2(2.25) + 4(1.5 × $\frac{4}{3}$) = 4.5 + 8 = 12.5 sq ft. B uses less.
7	If cardboard costs $0.40 per sq ft, how much is saved making 20 boxes of design B instead of A?	0.5 sq ft × 20 = 10 sq ft × 0.40 = $4.00
8	Explain why a cube-like box uses less cardboard for the same volume.	Shapes closer to a cube have less surface area for the same volume because no face is very long and thin.

Part 3: Pitch the Makeover

#	Task	Answer
9	A bookcase is a rectangular prism 3 ft wide, 1 ft deep, $5\frac{1}{2}$ ft tall with 4 equal shelves. What is the volume of each shelf space?	3 × 1 × 5.5 = 16.5 cubic ft ÷ 4 = 4.125 cubic ft per shelf
10	Choose the floor plan, rug, paint, boxes, and decals. Find the total cost and the percent of the budget used.	Answers vary; percent = total ÷ 900 × 100.
11	Write a recommendation for the TV show explaining one choice where you used geometry to save money.	Answers vary; look for a specific area, surface area, or volume comparison.

Reflection

#	Prompt	Look for
R1	How can two boxes have the same volume but different surface areas?	Different dimensions can multiply to the same volume while their face areas add to different totals.
R2	When decomposing a polygon, how do you make sure you do not count area twice?	Use pieces that do not overlap and that together cover the whole shape exactly.